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Core Grain Size Analysis Using a Handheld Device

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Abstract

This study presents a novel handheld device for automated grain size analysis in geological cores, integrating hardware, software, and machine learning to optimize traditional labor-intensive methods. The developed tool combines a high-resolution camera, touchscreen display, and single-board computer to capture and process core images in near real-time. A YOLOv9-based segmentation model, trained on a custom dataset augmented by the Segment Anything in High Quality (SamHQ) model, identifies individual grains, while contour extraction and ellipse fitting algorithms quantify grain morphology. Post-processing employs a gradient boosting model to correct biases in median grain size estimates, aligning results with expert interpretations. Field tests demonstrate the device's ability to generate accurate grain size distributions and depth profiles, with applications in reservoir characterization and lithological analysis. The solution offers a scalable, efficient alternative to manual core description, enhancing objectivity and reducing analysis time in both lab and field environments.

Introduction

While drilling a well for exploration or development, extracted cores are an important source of information needed for subsurface characterization. One of the methods of petrophysical research is the lithological study of the geological core, in particular the macroscopic description of the core and the construction of a lithological column of sediments. In this case, the diagnostic features of the core are color, structure, composition, cementation, texture, presence of fracture, visual emptiness, and others. Some analyses are performed before slabbing the core and others are done after the slabbing. One of the most labor-intensive, time-consuming and subjective processes is the analysis of the structure, which includes the study of the size and shape of grains and the presence of inclusions. Traditionally, using a magnifying glass the geologist visually inspects the cut slab and records the visual properties of the core according to her/his knowledge and experience. Existing automated approaches often depend on background separation techniques or machine learning methods tailored to narrow, specific tasks, limiting their flexibility and broader applicability. To ensure fast, effective, and accurate results it is necessary to automate the process of each step from image capturing up to result interpretation. To improve the efficiency and objectivity of this process, we propose

a novel portable laboratory tool – a compact handheld device designed to assist geologists in partially automating core sample description. Our integrated solution combines hardware, software, and machine learning into a single portable handheld device, offering a streamlined alternative to traditional lab-bound workflows. This prototype enables near real-time grain size estimation of cores through grain segmentation and advanced image processing techniques.

The first component of our solution is a custom designed device equipped with a high-resolution onboard camera and a lens system to provide detailed image capture of geological samples. A built-in display enables immediate visual feedback, and a powerful computational unit is integrated to support intensive processing tasks, including the execution of machine learning models. The second component focuses on the software developed specifically for core grain size estimation task. This software facilitates image capture, live observation, prediction of grain sizes, and analysis of results. The core of the image processing pipeline leverages a YOLOv9-based neural network, trained on a manually labeled dataset, to perform accurate grain segmentation. Following segmentation, the system extracts grain contours and analyzes their properties using dedicated algorithms. Post-processing is conducted using statistical distribution metrics to refine the final grain size estimation. The application features an intuitive graphical user interface (GUI), allowing users to capture and view images, manage historical data, and export results wirelessly.

The handheld prototype has been successfully tested in real laboratory conditions for both dataset collection and near real-time grain size analysis. Results indicate that the device delivers consistent and reliable performance with acceptable accuracy, demonstrating its potential as a practical tool for geologists and researchers in remote or in-lab environments.

Methods

Device overview

The key functionality of the developed device lies in its ability to capture high-resolution microscopic images of core surface and to identify and measure the size of individual grains within those images. To achieve this, the device integrates several key components: a digital camera, a precision lens system, a touch-screen, and a single-board computer (SBC), all housed within a custom-designed enclosure. A photograph of the assembled device is shown in [Fig. 1](#).



Figure 1—Photo of handheld device for core grain size estimation

Microscopic imaging is facilitated by the device's optical system, which includes a high-quality camera and a magnification lens. The optical system enables magnification levels of up to 200x, allowing for detailed observation of core textures and grain boundaries. A helicoid focusing mechanism provides manual control over image sharpness. Power light emitted diodes (LEDs) arranged in a circle ensures uniform illumination of the observed surface. Lighting can be adjusted by changing the power and turning off some of the LEDs. It is possible to control two LED channels independently and control light flow angle. The device's electrical architecture consists of three digital modules: control board, single-board computer (SBC) and camera. Communication between modules is carried out via serial digital interface. The SBC performs high-level of control: image processing, graphical user interface (GUI), and commands preparation for executive elements. To meet the substantial computational demands of real-time image processing, GUI rendering, and predictive modeling, the prototype employs an ARM64-based SBC. The control board implements low-level of control – handling of commands from high-level and light control. The control board is based on the STM32 microcontroller with a system of a light control unit and Battery Management System (BMS) module. The BMS module contains a step-up voltage converter, overcurrent protection, overcharge and over-discharge of batteries protection. The device is powered by an internal rechargeable battery, enabling portable, untethered operation for about 1 hour in laboratory environment.

Users can access real-time image previews, initiate machine learning predictions, and manage settings directly through the GUI. All control inputs are handled via the touchscreen, eliminating the need for external input devices. The integrated hardware design supports modular upgrades and future expansions, offering a scalable platform for broader petrophysical applications.

Device user control functions

Core analysis often involves evaluating grain sizes across hundreds of core sample images taken along the entire depth of a wellbore. A major challenge in this process is the need to maintain accurate associations between each image and its corresponding depth, as well as ensuring consistent naming for efficient data management. Additionally, certain samples may require higher magnification or resolution to capture fine details, further increasing operational complexity. Finally, the ability to export data for further analysis in external environments is essential. The device graphical user interface (GUI) was designed to optimize and automate these tasks in order to reduce manual workload and improve workflow efficiency.

The GUI was developed in Python for the Linux platform to provide convenient use. GUI provides full control over the device's core functionality through an integrated touchscreen display. The GUI is divided into four primary functional modules: setup page, metadata page, camera view page, and data export module. The GUI functional pages and key operation options are shown in [Fig. 2](#).

The set-up page provides advanced configuration options, allowing users to adjust image resolution, toggle automatic segmentation for newly captured images, and enable developer/debug mode for maintenance or troubleshooting. These settings offer flexibility to meet specific requirements, whether prioritizing speed, accuracy, or raw data collection. Metadata page enables users to input and manage metadata related to core sample depth. The interface allows manual input of the current analysis depth and the step increment to be used for subsequent images. The system automatically applies these depth values to each captured image, helping to ensure accurate and consistent documentation of the core profile. This functionality greatly reduces the time and effort required for post-processing and data organization. The camera view page is used for real-time inspection of core structures. When this mode is activated, the camera is initialized and the lighting system is automatically turned on. User can view a live image feed from the microscope and capture images with a single touch. The dedicated capture button saves the image and initiates the machine learning prediction pipeline simultaneously. Once processing is complete, the original image, along with the predicted grain segmentation data and associated statistics, is stored in the device's internal memory. Each image is linked to a specific depth, enabling automatic construction of a stratigraphic profile in a tabular format as new data is collected.

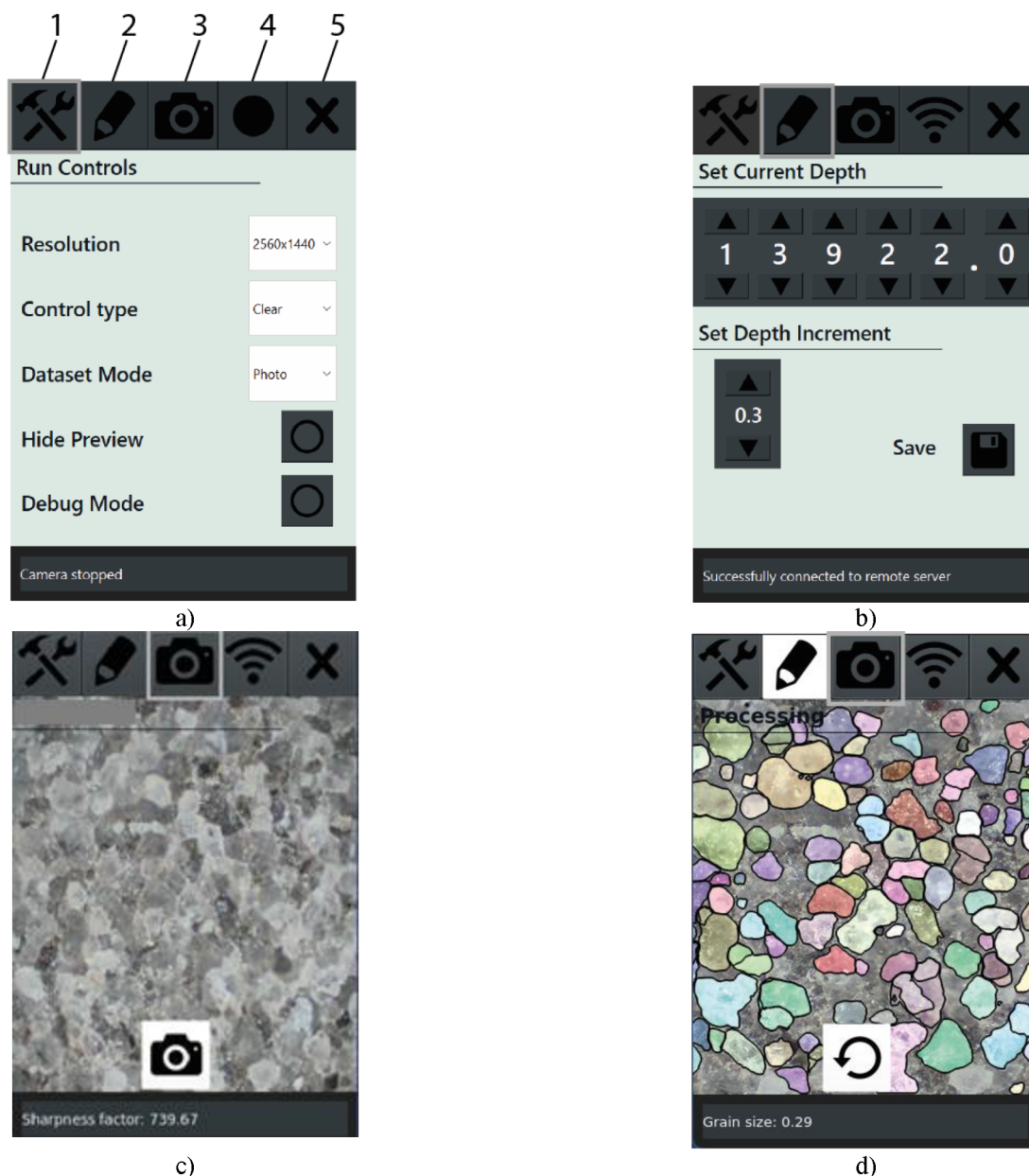


Figure 2—GUI functional pages and key operation options. Page navigation panel includes the following buttons: 1 – settings. 2 – metadata setup. 3 – camera view. 4 – data export button. 5 – close window (available in debug mode). a) Setup page allows to customize device operation logic; b) Metadata setup page for manual input of wellbore depth and depth increment configuration; c) Camera view page – real-time preview of core sample; d) Camera view page with displayed grain segmentation results and grain size estimation.

The data export module allows communication with external systems via a built-in Wi-Fi module. An external adapter program initiates and manages data transfer by requesting the desired datasets from the device. This wireless export mechanism simplifies the integration of the device into digital data workflows, allowing geologists to continue processing and interpreting the data on workstations or cloud-based platforms. Overall, the modular and user-friendly control system enhances the device's utility in both field and lab environments, making it a versatile tool for efficient and scalable geological core analysis.

Grain size estimation pipeline

Grain size estimation pipeline is shown in Fig. 3 and consists of several functional stages (tasks): photo grabbing, image preprocessing, segmentation model inference, segments processing, statistics collection, and grain postprocessing. The photo grabbing stage uses image stream logic to capture images. This

functional logic includes managing of image stream and camera parameters. For preview mode, it operates at a lower resolution to match the device display resolution and reopens at a higher resolution based on the device's manual setting during image capturing. Light system is configured to work only while camera stream is running. Image preprocessing includes adjusting the image scale and converting it to an appropriate color format. The captured image resolution is defined by an application settings parameter.

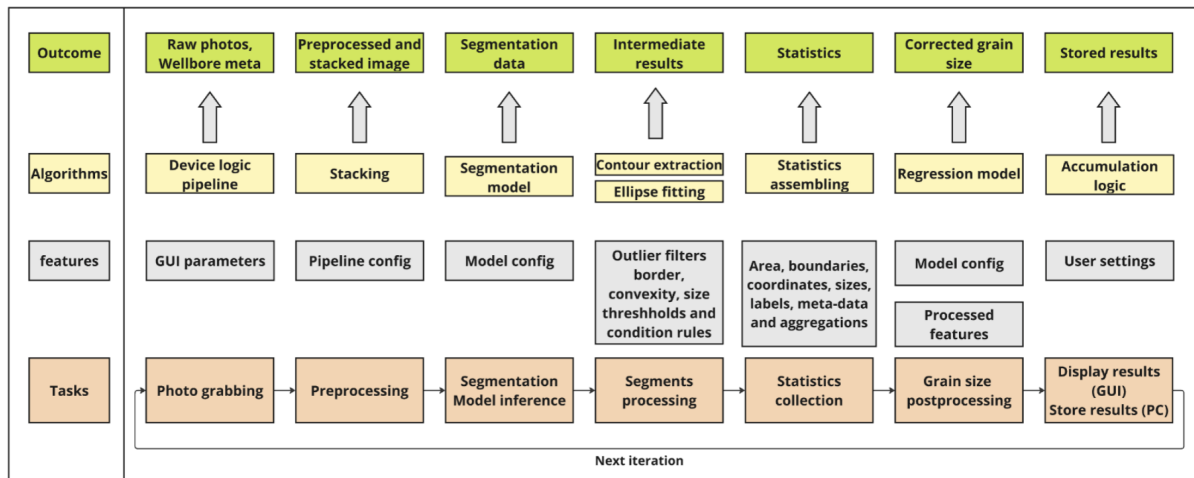


Figure 3—Device functional logic pipeline and data flow

Segmentation model inference is a stage where the image is being processed with segmentation ML model. We address segmentation as an instance segmentation task with grain instances. The Resulting segmentation data is represented by binary masks that contain information about which image pixels belong to a grain label class. The segmentation behavior can be adjusted by model configuration; some features can be changed from settings management page through user interaction. Segments processing includes grain contours extraction and analysis of its geometry. This process will be described in detail in a later section. The statistics collection stage is used to organize the features collected in previous steps, such as image depth, lens magnification, and detected grains, including their sizes, areas, and contour parameters. In this step, statistical calculations are also performed and saved to memory. Grain postprocessing concludes the processing engine, which include domain grains size estimation over grain distribution analysis and distribution shift refinement with gradient boosting algorithm.

Grain segmentation. The grain segmentation stage is a critical component of the overall workflow, since it involves analyzing each detected grain individually to estimate key geometric and morphological properties such as shape, area, and size. These metrics are essential for characterizing the textural attributes of cores and serve as input parameters for broader geological interpretations.

This process begins with image processing through the segmentation model based on Yolov9 network (Wang et al., 2024), trained on our own collected and labeled dataset. To support labeling process and reduce labor, we used intermediate segmentation model based on Segment Anything in High Quality model (SamHQ, Ke et al., 2023). SamHQ generates grain masks on its output as binary representations to highlight grain regions in the captured image. These masks are processed to extract the contours of individual grains. To achieve this, we apply a binary threshold to the mask to distinguish adjacent pixels belonging to different grains, effectively creating sharp boundaries between segmented areas.

The contours are extracted using OpenCV's implementation of the border-following algorithm, which is based on the Suzuki–Abe method (Suzuki et al., 1985). This algorithm identifies the continuous curve that defines the outer boundary of each grain, where all boundary points share the same intensity value. The result is a set of grain contours, each represented as a closed polygonal curve. To ensure accuracy and

eliminate noise, the extracted contours undergo a filtering stage. This stage discards irregular or fragmented contours that do not meet predefined criteria for size, shape, or compactness – features, which typically correspond to segmentation artifacts or non-grain features.

Once valid contours are identified, the next step involves extracting geometric parameters for each grain. This is done by fitting an ellipse to each contour on the image in a least-squares sense (Fitzgibbon, 1970). The fitted ellipse provides two key measurements: the major and minor axis lengths, which are used to calculate the equivalent grain diameter. Specifically, the average of the major and minor axis lengths is taken as the grain's representative size in pixels. In addition to size, the area enclosed by the contour is computed in pixels. To convert pixel-based measurements into millimeters, we apply a manually created calibration table. This table accounts for the specific optical configuration of the device, including the magnification level of the lens system and the pixel size of the image sensor.

A detailed overview of the grain contour extraction and ellipse fitting process is illustrated in Fig. 4. Example of grain size evaluation and its representation is shown in Fig. 5. This approach ensures consistent data collection of grain morphology, supporting further distribution analysis, statistical evaluation and results accumulation for each core sample.

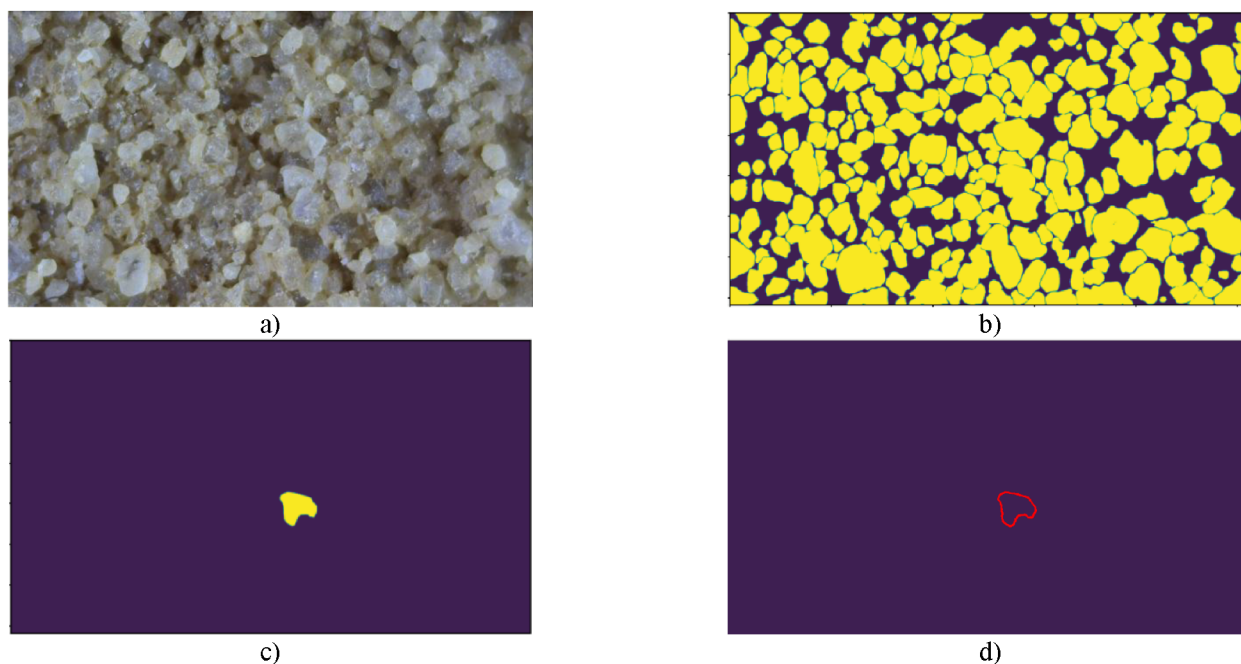


Figure 4—Grain segmentation masks processing transform: a) original image, b) predicted binary segmentation masks, c) single binary mask of a detected grain, d) grain binary mask contour

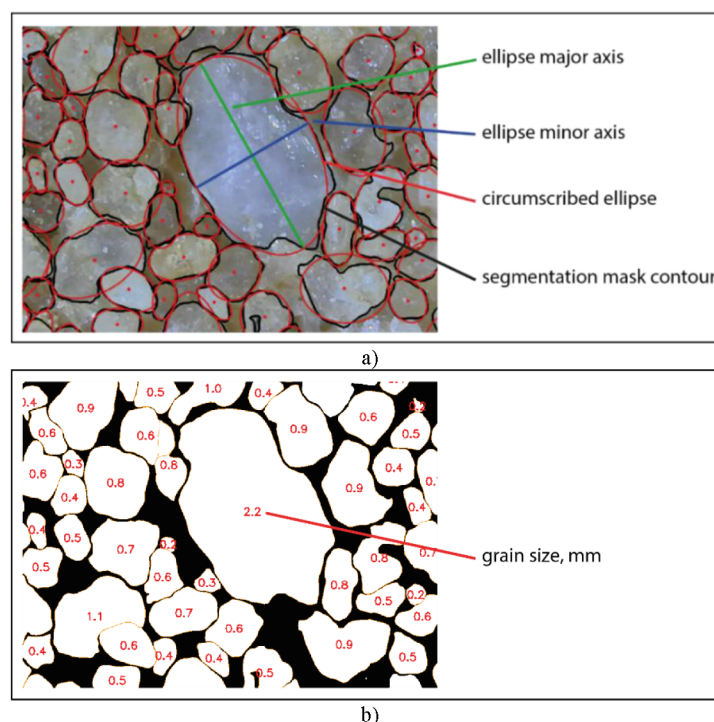


Figure 5—Grain size evaluation from contours results representation: a) image with mask contours circumscribed with ellipse; b) grain binary masks with correspondent grain sizes

Distribution processing and result grain size estimation. A single core image typically contains a large number of grains, often exceeding 200 distinct grains within a relatively small surface area of approximately 14 mm². To determine the effective grain size, we analyze the aggregated grain size distribution derived from segmentation processing results. By collecting aggregated information about grain size distribution of core at a particular depth geologists can determine the geological characteristics of the formation (Buscombe, 2008). The grain segmentation processing stage outputs individual grain measurements, which are then compiled into a statistical distribution. To improve accuracy and reduce variability in the results, multiple images from the same depth interval can be processed. This approach increases the sample size, thereby smoothing the overall distribution and reducing the impact of local anomalies or segmentation noise. An example of a resulting grain size density distribution plot is presented in Fig. 6.

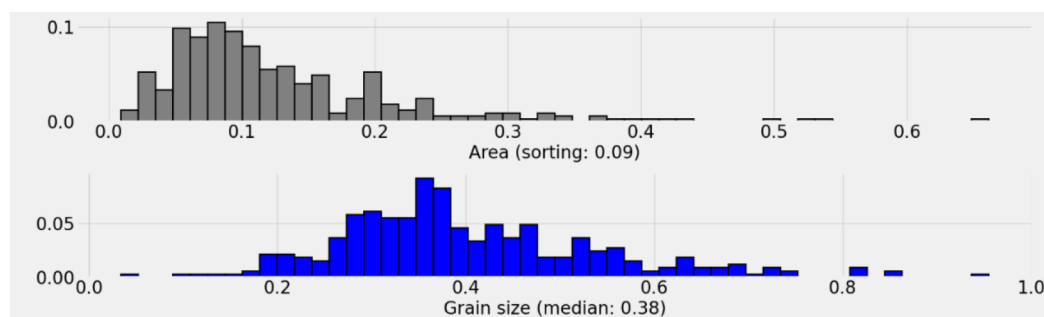


Figure 6—A density distribution plots for grain area and grain size parameters

The initial grain size estimation is determined by calculating the median of the detected grain sizes across the cores. The median is chosen for its robustness against outliers, particularly when dealing with skewed distributions. In practice, certain limitations arise. Cores can contain heterogeneous grain clusters where smaller grains dominate numerically, while larger grains, though geologically significant, are

underrepresented in count-based statistics. As a result, the median grain size may be biased toward smaller values, failing to reflect the true geological composition of the sample.

To address this limitation, we implemented an additional post-processing step utilizing a machine learning-based correction model. Specifically, we employed a gradient boosting algorithm trained on a dataset labeled with expert-corrected grain size values. These expert adjustments serve as ground truth references, allowing the model to learn the typical deviations between raw median estimates and geologically informed assessments. The input features for the model are derived from histogram bins of the grain size distribution, capturing the overall shape and balance of the distribution. The model predicts a correction factor, which is applied to the raw median value to obtain a more accurate and representative grain size estimate. This post-processing correction model functions as a data-driven expert system, effectively bridging the gap between automated analysis and human expertise. In validation experiments, the system demonstrated a significant improvement in aligning predicted grain sizes with those determined by experienced geologists. This hybrid approach enhances interpretability of grain size measurements, making the solution suited for practical deployment.

Experimental results

The primary objective of the developed device and its associated software pipeline is to generate a comprehensive grain size distribution profile across the entire depth interval of the target wellbore. This profile is constructed by aggregating grain size predictions and related statistical parameters obtained from the analysis of core images captured at various depths. In a typical workflow, hundreds of images are acquired along the depth of the borehole, each corresponding to a specific sampling interval. For every image, the device processes the data and extracts grain size metrics over segmentation and processing pipeline, as described in previous sections. These metrics include the estimated grain size, grain count, shape descriptors, and distribution statistics. All values are recorded in structured tabular form, with depth values ordered sequentially from the bottom to the top of the analyzed section. This accumulated dataset serves as a digital representation of the wellbore's lithological characteristics. By using tabulated data, geologists can interpret variations in grain size within depth range, which can be indicative of depositional environments, properties of a hydrocarbon reservoir, and formation heterogeneity.

To further enhance visualization and interpretability, the collected data by device can also be presented graphically as a continuous wellbore profile plot. In this format, grain size values are plotted against depth, producing a visual representation of textural changes within the core. Such profiles offer a quick and intuitive way to identify zones of interest, for instance transitions between fine-grained and coarse-grained layers, or intervals exhibiting anomalous grain distributions. The generated profiles can serve as a foundational layer for further geological interpretation, such as stratigraphic correlation, facies analysis, or reservoir property estimation. An example of an automatically generated profile is illustrated in [Fig. 7](#).

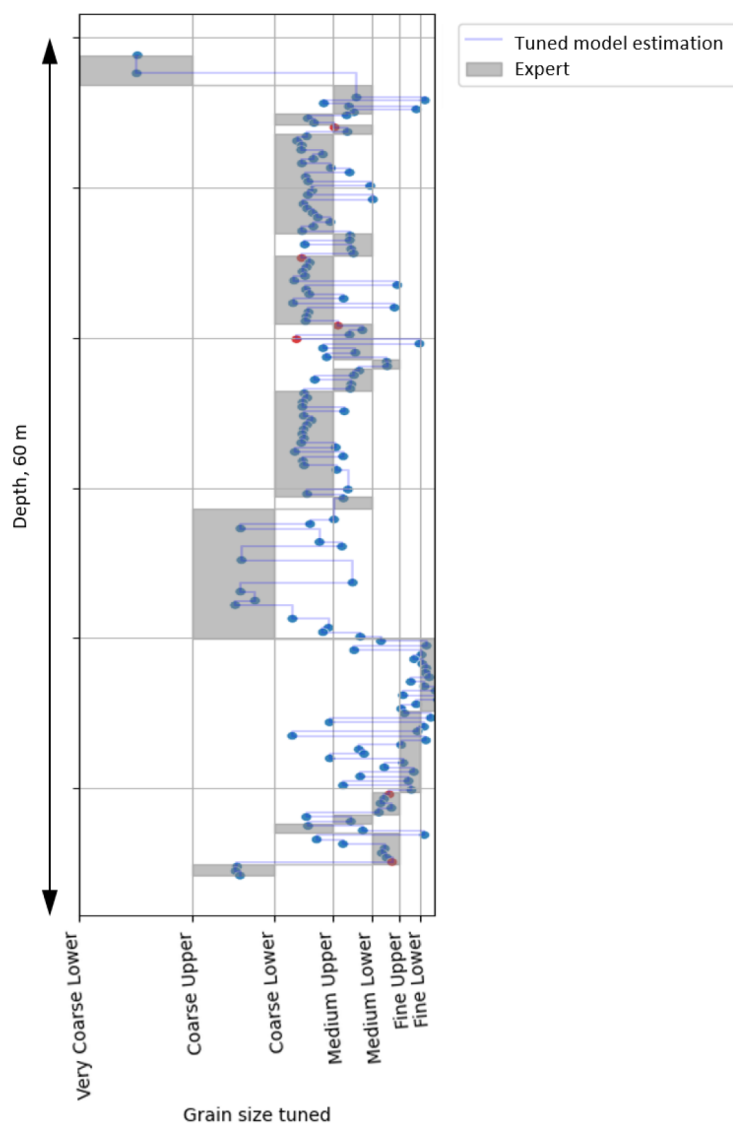


Figure 7—An example of aggregated plot for wellbore grain size profile (FOV4.8, accuracy: 65.19 %)

Conclusions

In this work, we presented a comprehensive and integrated approach for the problem of automatic grain size estimation of geological cores. Our approach combines hardware, software, image processing algorithms, and machine learning models into a single portable handheld device. The system is designed to assist geologists and ease the expansion of the traditionally manual and time-consuming process of core analysis. Throughout the development process, we addressed a number of challenges within mechanical, electrical and software areas. These included determining the optimal optical magnification to reliably capture a wide range of grain sizes, configuring and calibrating the lens and focusing system, and establishing efficient electrical and logical communication between all device components. Additionally, we designed and manufactured an ergonomic and compact device casing using 3D printing, ensuring ease of use in laboratory conditions. On the software side, we developed a GUI that enables intuitive control of the device, real-time image visualization, and access to analytical functions. The core image processing pipeline was implemented combining algorithmic approach with two machine learning models: one for grain segmentation based on YOLOv9, and another for post-processing corrections using gradient boosting. The dataset required for model training was prepared and labeled manually, with particular attention paid to challenging cases involving low contrast, overlapping grains, or complex rock textures.

The prototype is currently in the testing phase, where it has demonstrated good potential for application use. Initial tests confirmed the system's capability to produce reliable grain size estimations and generate continuous depth profiles suitable for further geological interpretation. It is important to emphasize that the prototype provides a flexible platform for future enhancements. Both hardware and software components can be further optimized to improve performance, user experience, and prediction accuracy. Additionally, the system architecture allows for expansion toward related analytical tasks, such as studying pore structure, mineral matrix composition, or automated lithofacies classification. Overall, the results confirm practical utility of the proposed device integrated system for automated core analysis, which can mitigate use of traditional manual methods. The authors share an opinion that proposed solution represents a step toward digitizing and automating geological core analysis, offering a comprehensive tool for grain size estimation task.

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